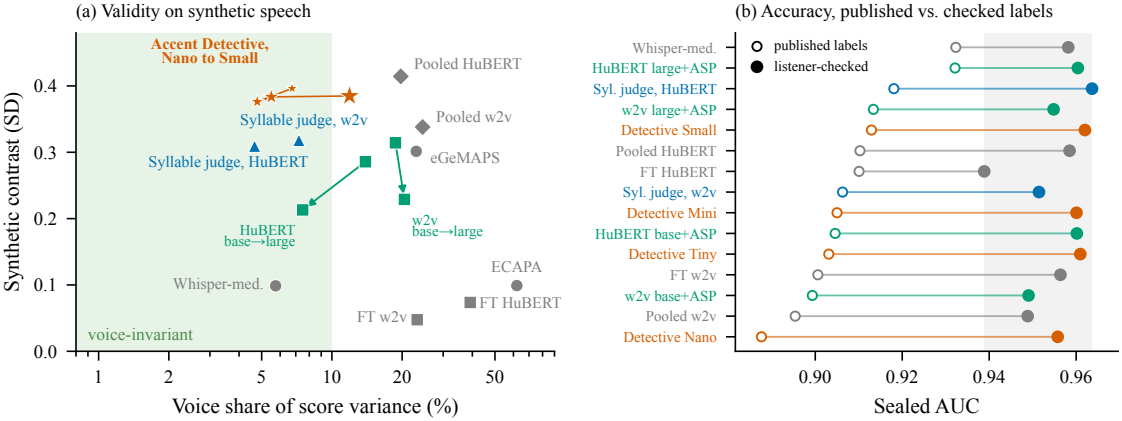

(a) Validity on synthetic speech
Accent Detective, Nano to Small
Pooled HuBERT
Pooled w2v
eGeMAPS
Syllable judge, w2v
Syllable judge, HuBERT
HuBERT base→large
w2v base→large
ECAPA
Whisper-med.
FT HuBERT
FT w2v
voice-invariant
Synthetic contrast (SD)
0.0
0.1
0.2
0.3
0.4
1
2
5
10
20
50
Voice share of score variance (%)
(b) Accuracy, published vs. checked labels
published labels
listener-checked
Whisper-med.
HuBERT large+ASP
Syl. judge, HuBERT
w2v large+ASP
Detective Small
Pooled HuBERT
FT HuBERT
Syl. judge, w2v
Detective Mini
HuBERT base+ASP
Detective Tiny
FT w2v
w2v base+ASP
Pooled w2v
Detective Nano
0.90
0.92
0.94
0.96
Sealed AUC